

Unit 1, Lesson 6: Writing and Solving Two-Step Equations

Benchmark

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Learning Targets

- ☐ I can solve a two-step equation within a mathematical or real-world context.
- ☐ I can write a two-step equation given a real-world context.

1.6.1 Warm-Up: Estimation

The picture shows 100 #4 size paper clips with four clips chained together.



1. Complete the statements by estimating how long it would take you to chain 100 paper clips together.
 - a. Too Low: _____
 - b. Too High: _____
 - c. Just Right: _____

1.6.2 Guided Practice: Properties of Equality

Properties of equality can be used to solve equations.

1. Complete the table with a numeric example of each property of equality.

Property	Algebraic Statement	Numeric Example
Symmetric Property of Equality	If $a = b$, then $b = a$.	
Addition Property of Equality	If $a = b$, then $a + c = b + c$.	
Subtraction Property of Equality	If $a = b$, then $a - c = b - c$.	
Multiplication Property of Equality	If $a = b$, then $a \cdot c = b \cdot c$.	
Division Property of Equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.	

2. Which property justifies the equation $4m + 7.1 = 24.38$ is equivalent to $24.38 = 4m + 7.1$?

3. Solve $-8.1 = \frac{x}{-3} - \frac{2}{5}z$ by applying properties of equality.

- a. Fill in each box with an operation and each blank with a value.

$$\begin{aligned}
 & -8.1 = \frac{x}{-3} - \frac{2}{5}z \\
 & -8.1 \boxed{} \text{---} = \frac{x}{-3} - \frac{2}{5} \boxed{} \text{---} \\
 & \text{---} = \frac{x}{-3} \\
 & \text{---} \boxed{} \text{---} = \text{---} \boxed{} \frac{x}{-3} \\
 & \text{---} = z
 \end{aligned}$$

- b. Use substitution in the original equation to verify the solution.

$$-8.1 = \frac{x}{-3} - \frac{2}{5}z$$

4. Solve $5x + 4.25 = -3\frac{1}{4}$ by applying properties of equality.

- a. Fill in each box with an operation and each blank with a value for the first step in solving the equation. Then, finish solving the equation on your own.

$$5x + 4.25 = -3\frac{1}{4}$$

$$5x + 4.25 \boxed{} = -3\frac{1}{4} \boxed{} $$

$$5x = $$

$$x = $$

- b. Use substitution in the original equation to verify the solution.

$$5x + 4.25 = -3\frac{1}{4}$$

1.6.3 Try It: Solving Two-Step Equations

1. Solve each equation by applying properties of equality. Use substitution to verify the solutions for each equation.

$$-4w - 7 = -11.2$$

$$\frac{g}{8} + \frac{1}{2} = 0$$

$$10.5 = \frac{r}{6} - \frac{3}{4}$$

$$-\frac{12}{5} = 5m + 8.2$$

1.6.4 Collaboration: Using the Distributive Property

Equations can be solved in more than one way. When an equation is written in the form $a(b + c) = d$, it can be solved by either distributing a first or by dividing both sides of the equation by a first.

- 1. The equation $63 = 3(6 - p)$ can be solved using either method.
 - a. Solve the equation $63 = 3(6 - p)$ using each method.

Method 1 Apply the distributive property first.	Method 2 Divide both sides by the factor being distributed first.
$63 = 3(6 - p)$	$63 = 3(6 - p)$

- b. Verify that the solution is correct using substitution.

1.6.5 Guided Instruction: Writing a Two-Step Equation from a Real-World Context

When writing equations to represent a real-world context, it is helpful to start by considering the known and unknown quantities involved in the situation.

1. Darius is draining his pool for resurfacing. The pool began with 13,650 gallons of water and it is draining at a rate of 640 gallons per hour. How long has the pool been draining if currently there are 9,330 gallons in the pool?
- a. Complete the table to describe the known and unknown quantities in this situation.

Known Quantities	Unknown Quantities

- b. With your partner, discuss how the quantities are related in this situation. Summarize your discussion.



When writing equations to represent real-world contexts, **any variables used must be defined** to identify which quantity each variable in the equation represents.

- c. **Define the unknown quantity using a variable.**

- d. **Write an equation that could be used to find the unknown quantity using the variable you defined.**

- e. **Solve the equation.**

- f. **Explain what the solution means in this context.**

1.6.7 Wrap-Up: Error Analysis

Two students solved an equation. One student made a mistake. Their work is shown below.

Caroline	Ivy
$\frac{1}{3}x + 7 = 40$	$\frac{1}{3}x + 7 = 40$
$3\left(\frac{1}{3}x + 7 = 40\right)$	$\frac{1}{3}x = 33$
$x + 21 = 120$	$x = 11$
$x = 99$	

1. Which student made an error when solving the equation?
2. Explain what error the student made.